Fiscal Year:	FY 2015	**Task Last Updated:**	FY 09/17/2015
PI Name:	Williams, Forman A. Ph.D.		
Project Title:	Flammability Limit Experiment (FLEX)		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	La Jolla	**State:**	CA
Zip Code:	92093-0411	**Congressional District:**	53
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	OTHER
Start Date:	08/01/2007	**End Date:**	07/31/2016
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA GRC
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Flight Program:	ISS		
Flight Assignment:	ISS		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Dryer, Frederick Ph.D. (Experiment Co-I: Princeton University) Choi, Mun Young Ph.D. (Experiment Co-I: University of Connecticut) Shaw, Benjamin Ph.D. (Experiment Co-I: University of California, Davis) Avedisian, C. Thomas Ph.D. (Experiment Co-I: Cornell University) Farouk, Tanvir Ph.D. (Experiment Co-I: University of South Carolina) Hicks, Michael M.S. (Experiment Co-I: NASA Glenn Research Center) Dietrich, Daniel Ph.D. (Experiment Co-I: NASA Glenn Research Center) Nayagam, Vedha Ph.D. (Experiment Co-I: Case Western Reserve University)		
Grant/Contract No.:	NNX09AM33A		
Performance Goal No.:			
Performance Goal Text:			

Task Description:

The overall objective of the FLEX experiments (Flammability Limit Experiment) is to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives are to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios.
The first phase of FLEX was a focused set of science objectives designed to address spacecraft fire safety concerns. The primary objectives were the assessment of flammability limits in a range of habitable atmospheres and fire suppressant efficacies. The second phase is FLEX-2 with a more expansive set of science objectives designed to investigate fundamental phenomena associated with (i) fuel sooting, (ii) development of reference fuels for fuel surrogates, (iii) low flow convective effects on diffusion flames, (iv) spacecraft fire safety, (v) droplet interactions in spray combustion, and (vi) condensed phase effects on burning using bi-component fuel blends.

The research team for FLEX comprises externally funded Principal Investigators (PIs) and internally funded PIs. The lead researcher is Forman Williams for both FLEX and FLEX-2 and can be considered the de facto “PI” for FLEX (i.e., the combined experiments). The externally funded PIs are grantees affiliated with universities. The internally funded PIs are Michael C. Hicks (NASA-Glenn Research Center-GRC), Daniel L. Dietrich (NASA-GRC), and Vedha Nayagam (Case Western Reserve University).

Flammability Limit Experiment (FLEX) Experiment PI: Forman Williams, Ph.D., University of California San Diego.

FLEX and FLEX-2 Experiment External CoInvestigators (CoIs):

Frederick Dryer, Ph.D., Princeton University. Task title: "Scientific Support for a Space Based Droplet Burning Experiment"

Mun Young Choi, Ph.D., University of Connecticut. Task title: "FLEX Droplet Combustion Experiments in Microgravity"

FLEX-2 Experiment External CoInvestigators (CoIs):

C. Thomas Avedisian, Ph.D., Cornell University. Task title: "Combustion of Surrogate Fuel Droplets in Microgravity"

Benjamin Shaw, Ph.D., University of California Davis. Task title: "Scientific Support for the FLEX-2 ISS Experiment"

Tanvir Farouk, Ph.D., University of South Carolina. Task title: "Mathematical Modeling and Simulation of Space ased Droplet Combustion – Scientific Support to FLEX and FLEX-2 Program"

FLEX and FLEX-2 Experiment Internal CoInvestigators (CoIs):

Michael C. Hicks, NASA Glenn Research Center.

Daniel L. Dietrich, NASA Glenn Research Center.

Vedha Nayagam, Ph.D., Case Western Reserve University.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Task Progress:

The overall objective of the FLEX experiments (Flammability Limit Experiment) is to improve fundamental knowledge of droplet burning and other combustion processes relevant to fire safety in manned space environments. Specific objectives are to investigate influences of low-gravity environments on flammability conditions of different fuels in different oxidizing atmospheres representative of fire-suppression scenarios. Experiments with hepane as the fuel and with methanol as the fuel are in progress aboard the International Space Station (ISS).
During the reporting period, there was extensive discussion about the specific tests that were performed on ISS. The chemistry of combustion of the fuel affects the combustion and extinction conditions. Research at University of California San Diego (UCSD) addressed problems of improving chemical-kinetic descriptions. In particular, extensive consideration of cool-flame chemistry in normal alkane droplet combustion was completed. This aspect of the research will be a major activity in future work.

The project, in general, involves basic flammability-limit investigations relevant to fire safety issues for manned space flight. In standard fire-suppression techniques, a suppressant jet is directed towards a fire or a smoldering region. In considering these techniques for space applications, it is necessary to analyze the effect of suppressant transport to both the near-field and far-field of the region of interest. Fire prevention can also be implemented by modifying the atmospheric composition of the environment so as to reduce the likelihood of a self-sustaining combustion event. Finally, the flammability properties of materials can also be modified by the addition of materials which suppress ignitability, enhance charring of surfaces, or chemically inhibit self-sustained burning. The research addressed these aspects of fire safety in space.

During the previous reporting period, special attention was paid to combustion in atmospheres enriched in carbon dioxide. Burning at both normal pressure and elevated pressures was studied, and detailed consideration was given to effects of radiation, which can be enhanced in such atmospheres. Atmospheres containing carbon dioxide are especially relevant to fire-safety issues of ISS because its fire-suppression system is based on the use of CO2 extinguishers. This research helped to improve our understanding of the physical and chemical mechanisms by which these extinguishers function under microgravity conditions.

During the present reporting period, the unexpected dual-mode phenomenon of alkane droplets combustion, observed for the first time in ISS, has been addressed intensively. Possible explanations have previously been developed and evaluated, and a written communication about this unexpected phenomenon has been prepared. This was published in Combustion and Flame Flame 159, 3583-3588 (2012) and continues to attract a lot of attention. Besides our Princeton group, we now have leading groups in Italy and in Germany studying this new discovery. A joint paper on this topic with the collaborators in Aachen, Germany has now appeared: G. Paczko, N. Peters, K . Seshadri, and FA Williams, The role of cool-flame chemistry in quasi-steady combustion and extinction of n-heptane droplets, Combustion Theory and Modelling 18 (2014) 515–531. By adding cool-flame chemistry to the descriptions of the chemical kinetics, we are beginning to be able to reproduce these previously unexpected results. In particular, at UCSD reduced chemistry has

	now begun to be derived for this new mode of droplet combustion.
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